

An Extended Reality Simulator for Pulse Palpation Training

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ABSTRACT

We present an iterative design of an Extended Reality simulator for pulse palpation training. The goal is to develop an immersive and realistic simulator that includes haptic, visual, and auditory feedback modes. Existing pulse simulators are often bulky expensive, and unsuitable for use as training tools. They are not designed to incorporate the pulse's auditory feedback, a critical component of continuous pulse monitoring. We describe findings from a preliminary user study that measures the Quality of Service (QoS), Quality of User Interaction (QUI), Quality of Experience (QoE) in the Audio-Haptic and Haptic-only simulation scenarios. The results show that audio-haptic simulators fared slightly better than the haptic-only simulators.

Index Terms: Extended Reality—Pulse Simulator—Haptic devices—Multimodal Feedback

1 INTRODUCTION

Haptic feedback aids in the acquisition of fine motor control, particularly in the medical field. Simulations of repetitive, hygienic, and immersive medical procedures can be created using realistic visualization and haptic feedback. Embodied interactions and embodied user interfaces assist in achieving invisible user interfaces, transforming simulators from a 2D computer screen to a 3D physical space, and facilitating virtual space navigation. In fully immersive environments, physical interactions are restricted, whereas mixed reality (MR) environments allow embodiment and user interfaces [5].

Advanced Training Life Support (ATLS) was created by the American College of Surgeons to teach trauma doctors a systematic approach to treating trauma patients. Trauma doctors must take the patient's pulse during the circulation step. A human's pulse is a pressure wave that expands and contracts the artery as the heart contracts and beats.

Pulse palpation as a diagnostic tool is straightforward, inexpensive, and time-saving, but essential. Palpating a pulse is a learned skill, and medical students must practice palpating and classifying a pulse under supervision due to the subjectivity of measuring pulse force. Traditional medical education increases medical errors.

Extended Reality (XR) allows students to assess and treat trauma patients in immersive, risk-free, replicable environments. To address research gaps, we added a haptic-based radial pulse component to the ATLS simulator [3]. To study how multimodal feedback affects medical procedure learning, our simulator represents pulse using haptic and auditory feedback. By integrating pulse parameters, the simulator provides a realistic training environment.

2 LITERATURE REVIEW

Human patient simulators (HPSs) started in the late 1960s and grew in the 1980s and 1990s. They are used to teach respiratory physi-

ology, cardiovascular hemodynamics, and advanced clinical skills [4]. Modern HPSs have palpable pulses, heart, breath, and bowel sounds, and an intravenous arm. For example, Multak et al. created an HPS simulating blood pressure and heart rate variations in the pulse [9].

Using hydraulic devices lets developers simulate components of the human circulation system. Liu et al. designed a low-power, intelligent pulse simulation comprising four modules, a mechanical transmission module, a control circuit module, a hydraulic system module, and a model. Koo et al. developed a simulator that uses Magneto-Rheological (MR) fluids to generate “two-peak” pulse waveform patterns for different age groups by adjusting augmentation index AI [6]. The Pulse Width Modulation (PWM) technique regulates MR fluid flow to simulate pulses. Compact pulsation simulators use motor-controlled cam-follower actuators. Yang and Hsieh built a pulse simulator with CAM actuation and DC motor control [12]. Their simulator prototype provided realistic pulse waveform haptic feedback for various pulse pressures and heartbeats.

Electromagnetic actuators can be used to create portable human pulse simulators for training but there is limited research on their use. Luo et al. showed that high-frequency Linear Resonance Actuator (LRA) vibrotactile stimuli called Hap-pulse can produce realistic pulse wave temporal profiles on the user's fingertips [7]. They demonstrated remote palpation teaching, and health information sharing and communication in a Virtual Environment (VE).

3 PROBLEM DEFINITION

Overall, research in the area of XR medical simulators has made giant strides toward building haptic simulators to help train medical students to measure pulse. However, these devices currently have limitations. Commercially available pulse training simulators based on mannequins [1] are expensive, costing around 500 USD, and could be more affordable.

A review of existing simulators reveals a variety of methodologies, including force feedback devices such as Phantom Omni and Falcon, modular hydraulic equipment, Human Patient Simulators, Mechanical systems, and intelligent materials such as MR Fluid and electromagnetic actuators. The haptic systems constructed employing mechanisms such as the Cam-Follower and electromagnetic actuators are inexpensive, portable, and appropriate for teaching. Scalability is one of the significant disadvantages of mechanical systems, as we would require a new CAM to mimic a new pulse for a Cam-based system.

Based on the above findings, we used electromagnetic actuators to develop an affordable haptic system for our work. We also determined that none of the simulators developed have implemented all the modalities of the human pulse. This is a major drawback of using these simulators as training tools. In order to improve the functionality of the simulator as a training tool, it is necessary to integrate the various feedback modalities on the human pulse. The findings from the review also highlight the limitations of the current systems and call for developing an XR testbed that can be used to train and assess the skills of medical students.

4 ITERATIVE DESIGN

Studies and research have been conducted to determine how the pulse's various parameters change with one another. Mathers et al. in their research demonstrates the effects of blood oxygen saturation

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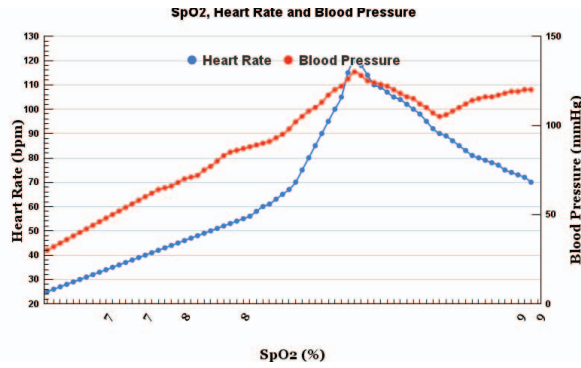


Figure 1: Variation in the pulse parameters simulated.

Table 1: Radial pulse parameter ranges for each category.

Category of Pulse	Heart Rate (bpm)	Blood Pressure (mmHg)	SpO ₂ (%)
Strong Pulse	95–120	120–130	90–95
Normal Pulse	65–95	100–120	95–99
Weak Pulse	45–65	70–100	85–90
Thready Pulse	35–45	50–70	80–85
No Pulse	20–35	30–50	70–80

on the electrocardiogram, blood pressure, and heart rate during the Anoxemia test [8]. We used this work when designing our simulator. For the pulse simulation, an oxygen-depleting trauma patient scenario was developed. In our scenario, the SpO₂ drops from 99 (maximum) to 70 and the corresponding heart rate and blood pressure are modelled accordingly. The heart rate ranges between 120 and 25 beats per minute (bpm). The blood pressure varies between 130 and 30 mm Hg.

Tachycardia and Bradycardia are classifications based on pulse rate. Pulses are classified according to its perceived force into strong/bounding, normal, weak, thready, and no pulse. The simulator's implementation focuses on developing hardware and simulating pulses of these categories. Table 1 shows the range of pulse parameters for each category. It takes practice to correctly classify the pulse into one of five categories by combining auditory and haptic feedback. Figure 1 displays the oxygen saturation, heart rates, and blood pressure of the simulated pulses. The transition between pulses is intended to produce a realistic simulation of a trauma situation.

There are three feedback modalities of pulse feedback: Haptic, Visual and Auditory. The auditory feedback provides information about the oxygen saturation of the pulse. The pitch of the audio from the multi-parameter model is representative of the oxygen saturation. The haptic feedback helps a trainee in learning about the pulse's force. For each feedback modality we describe the represented pulse parameters and the implementation.

The simulator consists of three components: the simulator software (a Microsoft HoloLens 2 app), the simulator hardware (a voice coil actuator controlled by Raspberry Pi4), and the network (Message Queuing Telemetry Transport (MQTT) server). The hardware component is responsible for providing the users with haptic pulse feedback. The simulator component provides the visual and auditory feedback of the radial pulse. The network component helps sync the auditory and visual feedback with the haptic feedback. Figure 2 represents the system design and architecture of the developed pulse simulator with its various components.

4.1 Haptic Design

Breitschaft et al. developed a haptic fidelity framework to categorize haptic technologies according to user experiences and mechanical

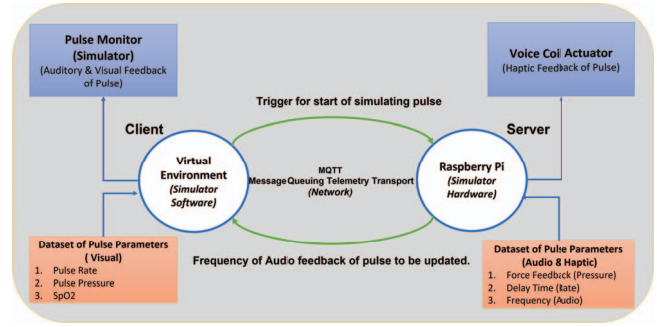


Figure 2: System Architecture of the pulse simulator.

properties [2]. Their work classified pulses into technologies with low, medium, and high haptic fidelity. Eccentric Rotation Mass (ERM) actuators are low-fidelity devices due to their low bandwidth, slow response, and longer haptic transients. Voice Coil Actuators (VCAs), defined by the precise presentation of a physical mechanism or mechanical user interface control, constitute medium fidelity devices. Magneto-Rheological Fluids and friction-modulating electrostatic materials are high-fidelity devices.

Our simulation uses VCA, a medium fidelity device. The simulator requires that we simulate pulses with varying strengths/forces, which demands feedback control of the actuator. Our haptic component is designed with a VCA developed by the Indian Institute of Technology, Madras. The initial configuration consists of a Raspberry Pi4, DAC MCP4725, Class D Power Amplifier, and voltage-regulated power supply as depicted in Figure 3. The actuator's cover design was inspired by the radial pulse procedure. We utilize the Gaussian approximation of pulse flow to actuate the radial pulse.

To design the haptic feedback, the force felt by a user was determined based on their blood pressure by assuming a tenfold attenuation factor. The simulated range of forces was between 0.3 N and 0.8 N. To achieve the above force range, we actuate the VCA at 8 V. The difference between the force feedback of any two simulation pulses is greater than the index finger's haptic force Just Noticeable Difference (JND). Using a Force Sensitive Resistor 04 Sensor, we validated the force experienced when the VCA is activated.

4.2 Auditory Design

Nellcor Corporation released in 1983 a pulse oximeter that correlated oxygen saturation with audio pulse tone pitch. A decrease in audio pitch indicates a decrease in the amount of oxygen reaching peripheral tissues, whereas an increase in audio pitch indicates an increase in tissue oxygen saturation. Expert trauma surgeons and anesthesiologists use audio pitch to monitor the patient's oxygenation and heart rate. This decreases the need for continuous visual monitoring of the display monitor. We calculate the audio frequency based on the patient's SpO₂, heart rate, and blood pressure using [11]. The audio feedback of the simulated pulses was designed using the JND of the audio frequency, which is 3 Hz for frequencies below 1000Hz. The simulated audio frequency range is 350 Hz to 750 Hz, and any two frequencies differ by more than 3 Hz.

The audio feedback is implemented by generating a sine wave of a specific frequency. A hardware component's MQTT message generates and plays the audio for a particular frequency. Sound spectrum analyzers confirmed the frequency of audio signals. Figure 4 shows a comparison of perceived sound to produced sound, with an average error of 10%. This error was compensated for when generating the required audio frequency.

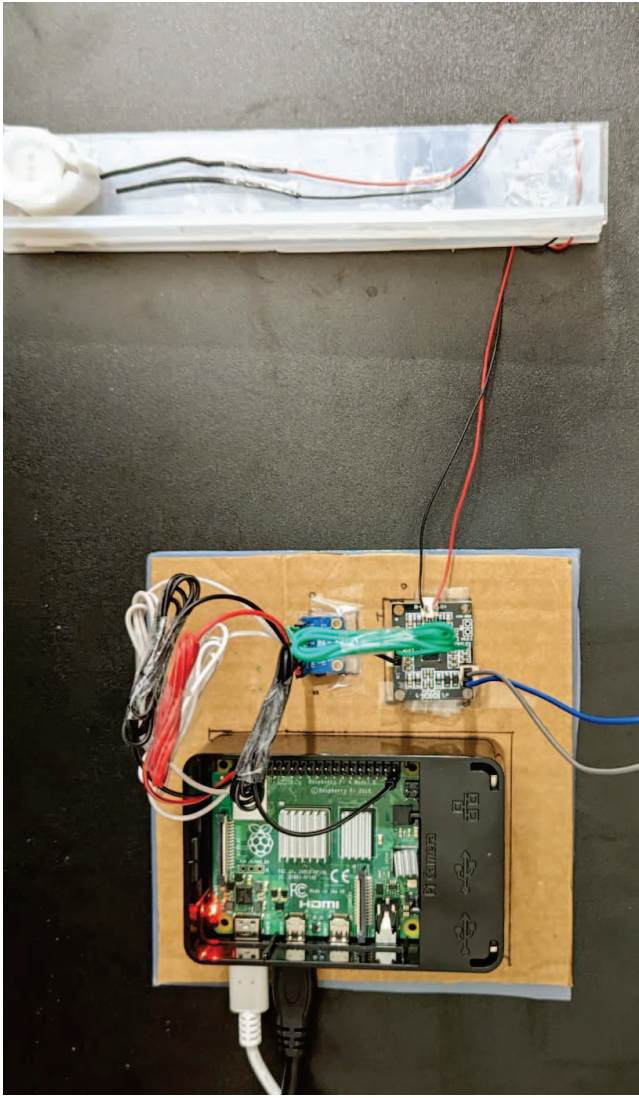


Figure 3: Hardware Design of the Pulse Simulator

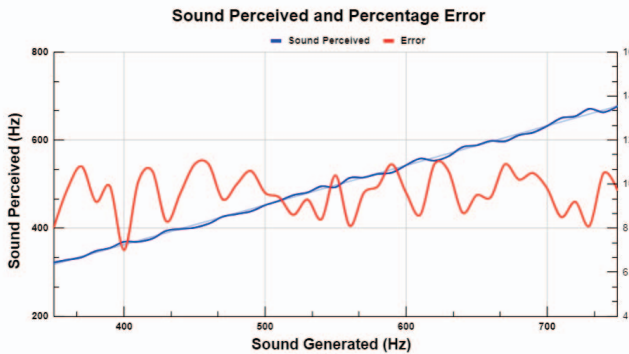


Figure 4: Validation of the sound generated by the simulator.

4.3 Integration with XR Simulator

The visual feedback of the pulse is the vitals monitor present in the scene that displays the values for the patient's vitals. Various de-



Figure 5: Multi-parameter display in the simulator providing visual feedback of the pulse.

signs of multi-parameter vitals monitors are available commercially, including a 4-parameter monitor and a 7-parameter monitor. The design of the vitals monitor in the XR simulator is minimalist and based on the commercially available 4-parameter monitor used in a trauma bay, which helps maintain the simulator's realism.

The monitor displays four parameters: Heart Rate (bpm), Blood Pressure (mmHg), Oxygen Saturation (percent), and Respiratory Rate. The monitor asset is updated once every second, independent of the haptic and audio feedback. It takes its value from the dataset of pulse parameters created (Figure 2).

The developed system integrates all the components using the MQTT message transfer protocol that provides a two-way communication channel. Small-sized string and integer values are transferred for the system to function.

5 PRELIMINARY DESIGN EVALUATION

Adding haptics and audio to VEs adds more realism and excitement for the users. Our preliminary study evaluates the simulator on various parameters. The pilot study aims to identify design flaws in the simulator and demonstrate their effect on the overall user experience. The evaluation metrics are a combination of subjective and objective metrics. Quality of Experience (QoE) for our study is defined as a linear combination of Quality of Service (QoS) and Quality of User Interaction (QUI) (Equation 1).

$$QoE = \alpha * QoS + (1 - \alpha) * QUI \quad (1)$$

In our case $\alpha = 0.4$. The objective of QoS parameters is to ensure the user experiences a seamless application flow. We have identified four parameters to define our prototype's QoS. The overall QoS is a linear combination of these four parameters (Equation 2).

$$QoS = a * L_{norm} + b * D_{norm} + c * AF_{norm} + d * HFF_{norm} \quad (2)$$

where $a + b + c + d = 1$. The four parameters are:

1. **Latency (L):** The latency between the message being sent to play the audio and the user perceiving the audio.
2. **Delay (D):** The delay in establishing all connections and starting the simulation
3. **Audio Frequency (AF):** The percentage error in the perceived audio frequencies
4. **Haptic Force Feedback (HFF):** The percentage error in the generated haptic force feedback

However, in our case, $a = b = c = d = 0.25$ as all four parameters significantly impact the user's perception and experience while using the simulator. The subjective metrics focuses on interpreting the user's experience and for our study it includes:

Table 2: Summary of the results for pilot study in the two scenarios.

	Haptic	Audio-Haptic
Quality of Service (QoS)	0.54 ± 0.07	0.45 ± 0.1
Quality of User Interaction (QUI)	0.56 ± 0.04	0.59 ± 0.3
Quality of Experience (QoE)	0.53 ± 0.07	0.57 ± 0.05

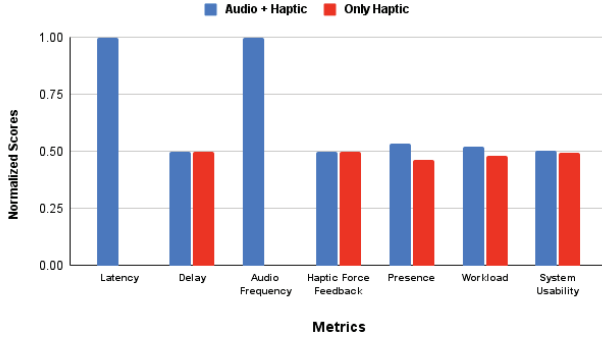


Figure 6: Average score values for QoS and QUI.

1. **System Usability (SU):** The system's usability will be collected using the standard System Usability Survey.
2. **User Presence (UP):** Presence perceived by the user in the VE is collected using the iGroup presence survey.
3. **User Task Workload (UTW):** The workload experienced by a user is measured using the NASA TLX questionnaire.

Therefore, QUI is (Equation 3):

$$QUI = p * SU_{norm} + q * UP_{norm} + r * UTW_{norm} \quad (3)$$

where $p + q + r = 1$ In our case, $p = q = r = \frac{1}{3}$ as all three subjective metrics are equally important and significant.

We performed a pilot study with ten participants (five male and five female). The study was a comparative study between audio-haptic enabled scenario and haptic only enabled scenario. In the study all the participants experienced a simulation with the pulses gradually changing from normal to absent. Five participants performed with Haptic Scenario, and the other five participants performed the study using the Audio-Haptic Scenario. The results for QoS, QUI, and QoE are summarized in Table 2. The ANOVA analysis shows that the improvement in the participants' experience is not statistically significant.

Figure 6 shows the average values of each the metric for QoS in both cases. The major design drawbacks that were identified are:

1. **No Spatial Mapping:** The scene rendering did not consider the positioning of the hardware. This negatively impacted the participant's simulator experience.
2. **Latency in Audio Generation:** Following the implementation of the Audio feedback, there was a perceivable lag between the haptic feedback and auditory feedback, which confused the users and increased the workload.

In the second design iteration we generated the Audio Feedback using a modified method (Section 6). We also added the spatial mapping feature to the simulator, which renders the patient's hand on the hardware simulator and builds the room size. These modifications addressed the significant issues identified in the pilot study.

6 SECOND DESIGN ITERATION

Due to the significant latency in the audio generation, we utilize a different technique to implement the audible feedback of the pulse.

This technique is based on the relationship between frequency and pitch. Frequency is the physical parameter of the sound, whereas pitch is what an individual perceives. Equation 4 provides the relationship between pitch and frequency [10]:

$$Pitch = 2595 \log_{10} \left(\frac{Frequency}{700} + 1 \right) \quad (4)$$

The pitch is measured on a logarithmic *mel* scale. In this technique, the pulse's audio feedback is simulated by calculating the pitch corresponding to the audio frequency and directly modifying the audio pitch. When a MQTT message is received from the hardware, the pitch of the audio is calculated using the above formula. After this, the pitch of the audio is modified and played accordingly.

We implemented the spatial mapping of the simulator using the capabilities of the Microsoft HoloLens 2-compatible Mixed Reality Toolkit (MRTK). Due to the addition of spatial mapping to the simulator in the second iteration, the patient's wrist is rendered on the hardware simulator and the room is built to room size. As a result, when the participant touches the patient's wrist to investigate the radial pulse, they will experience the haptic pulse feedback. These modifications addressed the major issues identified in the pilot study.

7 DISCUSSION & CONCLUSION

The medical field demands accurate and precise procedures to be performed by doctors, with no room for error. Traditional training methods need to provide the students with the required experience. Most of the existing research focuses on developing simulators with either haptic or auditory feedback. Doctors in the trauma room are required to continuously monitor the patient's pulse using the auditory and haptic feedback of the pulse.

Additionally, several pulse simulators were developed in the literature using various implementation techniques, but none provided auditory feedback. This is a research gap that is addressed in this work. For this work, we designed and developed an XR simulator on Microsoft HoloLens 2 to help train medical students about the palpation of the human pulse. We also combined all three modalities of feedback of the human pulse, which includes visual, audio and haptic. The result from the comparative evaluation of the XR training simulator is an essential first step toward understanding the advantages and limitations of the simulator for pulse feedback.

The major drawback of the pulse simulator system is that the rhythm of the pulse is not considered for the force feedback simulation. The rate and force experienced are the only two parameters considered for the haptic pulse feedback. We plan to perform a more robust study to understand the impact of multimodal feedback on learnability. Overall, this work provides a foundation for developing and demonstrating the utility of an XR multimodal simulator for pulse palpation training.

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